- RAG Learning

# 2.4 Graphs

OCR GCSE Mathematics (J560) — Paper 1, 2 & 3 — Foundation and Higher

## 2.4.1 Coordinates and Midpoints

A point in 2D is given by its **coordinates** $(x, y)$, where $x$ is the horizontal displacement from the origin and $y$ is the vertical displacement.

The **midpoint** of the line segment joining $(x_1, y_1)$ and $(x_2, y_2)$ is:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

The **distance** between two points (by Pythagoras):

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

**Worked Example:** Find the midpoint and length of the line segment joining $A(1, 3)$ and $B(7, 11)$.

> Midpoint: $M = \left(\frac{1+7}{2}, \frac{3+11}{2}\right) = (4, 7)$
>
> Length: $d = \sqrt{(7-1)^2 + (11-3)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$

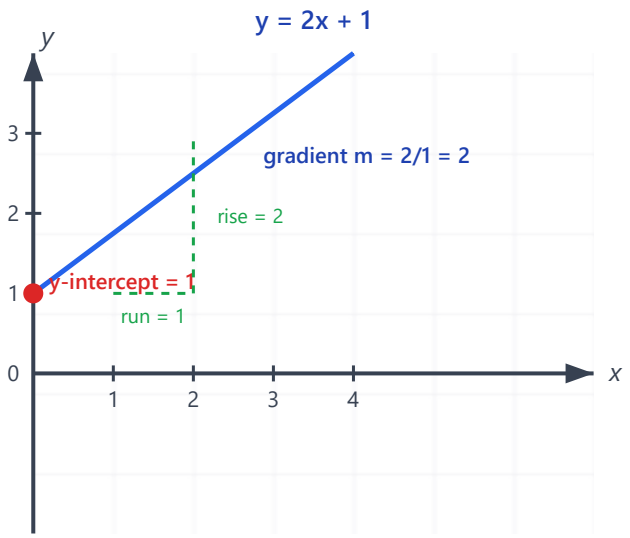
**Exam Tip:** The midpoint formula is an average of the two $x$-coordinates and an average of the two $y$-coordinates. You can think of it as "meet in the middle."

# 2.4.2 Straight Line Graphs

Any straight line can be written in the form $y = mx + c$, where:

- $m$ = gradient (slope): how steep the line is = $\frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x}$
- $c$ = $y$-intercept: where the line crosses the $y$-axis.

## Gradient and Equation of a Straight Line




**Worked Example:** Find the equation of the line through $(1, 3)$ and $(4, 9)$.

> Gradient: $m = \frac{9-3}{4-1} = \frac{6}{3} = 2$
>
> Using $y - y_1 = m(x - x_1)$ with point $(1, 3)$: $y - 3 = 2(x - 1)$
>
> Expand: $y = 2x - 2 + 3 = 2x + 1$

**Worked Example:** A line has equation $3y - 6x = 12$. State the gradient and $y$-intercept.

> Rearrange into $y = mx + c$ form: $3y = 6x + 12 \Rightarrow y = 2x + 4$

Gradient $m = 2$, $y$-intercept $= 4$ (i.e. the line crosses at $(0, 4)$).

### Higher — Parallel and Perpendicular Lines

- **Parallel lines** have the same gradient: $y = 3x + 1$ and $y = 3x - 5$ are parallel (both have $m = 3$).
- **Perpendicular lines** have gradients that are negative reciprocals: if line 1 has gradient $m$, then any perpendicular line has gradient $-\frac{1}{m}$. Their product is $-1$: $m_1 \times m_2 = -1$.

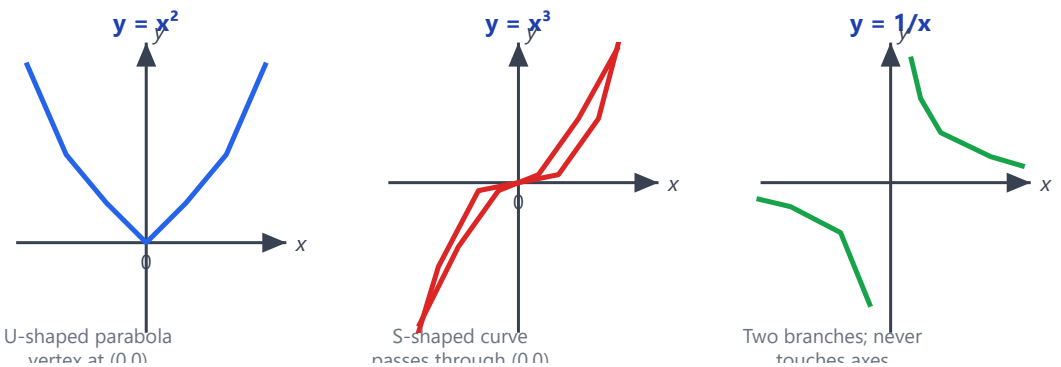
**Worked Example:** Find the equation of the line perpendicular to $y = 3x - 2$ that passes through $(6, 1)$.

Gradient of given line $= 3$. Perpendicular gradient $= -\frac{1}{3}$.

$$y - 1 = -\frac{1}{3}(x - 6)$$

$$y = -\frac{1}{3}x + 2 + 1 = -\frac{1}{3}x + 3$$

## 2.4.3 Quadratic, Cubic and Reciprocal Graphs




### Key Features of Each Curve

| Function | Shape | Key features |
|---|---|---|

| | | |
|---|---|---|
| $y = x^2$ | Parabola (U-shape) | Minimum at $(0, 0)$; symmetric about $y$-axis; always $\geq 0$ |
| $y = -x^2$ | Parabola (n-shape) | Maximum at $(0, 0)$; always $\leq 0$ |
| $y = x^3$ | S-shaped cubic | Passes through $(0, 0)$, $(-1, -1)$, $(1, 1)$; rotational symmetry about origin |
| $y = \frac{1}{x}$ | Reciprocal (hyperbola) | Two branches; asymptotes along both axes ($x = 0$ and $y = 0$); undefined at $x = 0$ |

### Plotting Quadratic Graphs: Table of Values

**Worked Example:** Plot $y = x^2 - 3x + 2$ for $-1 \leq x \leq 4$.

| $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|
| $x^2$ | 1 | 0 | 1 | 4 | 9 | 16 |
| $-3x$ | 3 | 0 | $-3$ | $-6$ | $-9$ | $-12$ |
| $+2$ | 2 | 2 | 2 | 2 | 2 | 2 |
| $y$ | **6** | **2** | **0** | **0** | **2** | **6** |

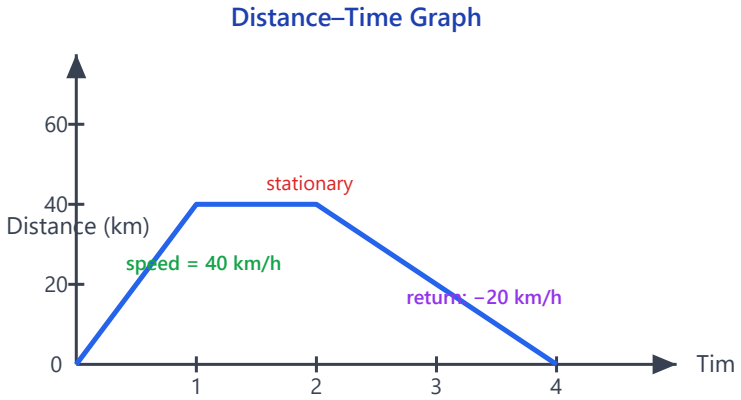
> $y$-intercept: $(0, 2)$. $x$-intercepts (roots): $(1, 0)$ and $(2, 0)$.
>
> Minimum between $x = 1$ and $x = 2$, at $x = 1.5$ (axis of symmetry).

**Exam Tip:** Always draw a smooth curve through plotted points — do not join with straight line segments. Plot at least 7 points for a quadratic.

## 2.4.4 Real-Life Graphs

## Distance–Time Graphs

- **Gradient** of a distance–time graph = **speed** $\left(\text{speed} = \dfrac{\text{distance}}{\text{time}}\right)$.
- A horizontal line means the object is **stationary**.
- A steeper line means greater speed.
- A negative gradient means moving *back towards the start*.




## Speed–Time Graphs

- **Gradient** of a speed–time graph = **acceleration** (in m/s$^2$).
- A horizontal line means **constant speed** (zero acceleration).
- **Area under the graph** = **distance travelled**.

**Worked Example:** A car accelerates from 0 to 20 m/s in 5 seconds, travels at 20 m/s for 10 seconds, then decelerates to 0 in 5 seconds. Find the total distance.

> The graph is a trapezium with parallel sides 10 s and 20 s, height 20 m/s.
>
> Area = $\frac{1}{2}(10 + 20) \times 20 = \frac{1}{2} \times 30 \times 20 = 300$ m

## Conversion Graphs

Straight-line graphs that convert between units (e.g. miles to km, £ to €). Read off values by locating one axis and reading across to the line, then down/up to the other axis.

**Exam Tip:** On distance–time graphs: steeper = faster. On speed–time graphs: steeper = greater acceleration. Area under speed–time graph = distance — and area under a velocity–time graph includes direction.

## 2.4.5 Transformations of Graphs [Higher]

### Higher only

If $y = f(x)$ is the original graph, the transformed graphs are:

| Transformation | Equation | Effect on graph |
|---|---|---|
| Translation up by $a$ | $y = f(x) + a$ | Shift the graph $a$ units **up** |
| Translation down by $a$ | $y = f(x) - a$ | Shift the graph $a$ units **down** |
| Translation left by $a$ | $y = f(x + a)$ | Shift the graph $a$ units **left** (counter-intuitive!) |
| Translation right by $a$ | $y = f(x - a)$ | Shift the graph $a$ units **right** |
| Reflection in $x$-axis | $y = -f(x)$ | Reflect in the $x$-axis (negate all $y$ values) |
| Reflection in $y$-axis | $y = f(-x)$ | Reflect in the $y$-axis (negate all $x$ values) |
| Stretch (vertical), scale factor $a$ | $y = af(x)$ | Multiply all $y$-coordinates by $a$ |
| Stretch (horizontal), scale factor $\frac{1}{a}$ | $y = f(ax)$ | Multiply all $x$-coordinates by $\frac{1}{a}$ (compress if $a > 1$) |

**Memory aid for horizontal shifts:** $y = f(x + a)$ shifts LEFT by $a$ because the same $y$-value now occurs at a smaller $x$-value. Think "inside the bracket does the opposite."

**Worked Example:** The graph of $y = f(x)$ passes through $(2, 5)$. State the coordinates of the corresponding point on: (a) $y = f(x) + 3$; (b) $y = f(x - 4)$; (c) $y = -f(x)$.

(a) $y = f(x) + 3$: add 3 to $y$-coordinate: $(2, 8)$

(b) $y = f(x - 4)$: add 4 to $x$-coordinate (shift right): $(6, 5)$

(c) $y = -f(x)$: negate $y$-coordinate: $(2, -5)$

**Exam Tip:** Transformation questions often ask you to apply a transformation to a sketch or to state coordinates of key points after transformation. Always think about what happens to a single point before drawing.

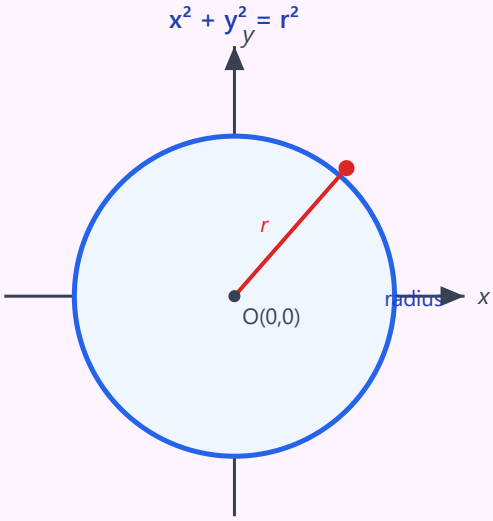
## 2.4.6 Equation of a Circle [Higher]

### Higher only

A circle with **centre at the origin** $(0, 0)$ and radius $r$ has equation:

$$x^2 + y^2 = r^2$$

This follows directly from Pythagoras' theorem: for any point $(x, y)$ on the circle, the distance to the centre is $\sqrt{x^2 + y^2} = r$.




**Worked Example:** (a) Write the equation of the circle with centre $(0, 0)$ and radius 7.
(b) Does the point $(3, \sqrt{40})$ lie on, inside, or outside this circle?

(a) $x^2 + y^2 = 49$

(b) Compute $3^2 + (\sqrt{40})^2 = 9 + 40 = 49$. Since this equals $r^2 = 49$, the point lies **on** the circle.

## Tangent to a Circle at a Point

A tangent to a circle at a point $P$ is perpendicular to the radius $OP$ at that point.

**Worked Example:** Find the equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

Gradient of radius $OP$: $m_{OP} = \dfrac{4-0}{3-0} = \dfrac{4}{3}$

Gradient of tangent (perpendicular): $m_t = -\dfrac{3}{4}$

Equation using point $(3, 4)$: $y - 4 = -\dfrac{3}{4}(x - 3)$

$y = -\dfrac{3}{4}x + \dfrac{9}{4} + 4 = -\dfrac{3}{4}x + \dfrac{25}{4}$

> Or in integer form: $3x + 4y = 25$

**Exam Tip:** To verify a point is on a circle: substitute its coordinates into the equation and check. To find the tangent: find the gradient of the radius, then take the negative reciprocal.

## Quick Reference Summary

| Topic | Key formula / fact |
|---|---|
| Midpoint of $(x_1, y_1)$ and $(x_2, y_2)$ | $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ |
| Gradient of line | $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ |
| Straight line equation | $y = mx + c$; or $y - y_1 = m(x - x_1)$ |
| Parallel lines [H] | Same gradient $m$ |
| Perpendicular lines [H] | Gradients satisfy $m_1 \times m_2 = -1$ |
| Distance–time graph | Gradient = speed; horizontal = stationary |
| Speed–time graph | Gradient = acceleration; area under = distance |
| $y = f(x) + a$ [H] | Translate up by $a$ |
| $y = f(x + a)$ [H] | Translate left by $a$ |
| $y = -f(x)$ [H] | Reflect in $x$-axis |
| Circle (centre origin) [H] | $x^2 + y^2 = r^2$ |
| Tangent to circle [H] | Perpendicular to radius at the point of contact |

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